

Health Facility Assessment (HFA) for small and sick newborns

Biomedical Technician Workshop Module

Version: February 2, 2023

Date (yyyy/mm/dd): / /

Data Collector 1 (first and last name):

Data Collector 2 (first and last name):

Disclaimer: This is a NEST360 | UNICEF global public good. When using content from this document, please include the following reference: “*This document includes content adapted from HFA Biomedical Technician Workshop Module, NEST360 | UNICEF, version 2023.02.02.*” The original NEST360 | UNICEF resources are located here: [NEST360.org/hfa](https://www.nest360.org/hfa).

Facility Name:

Facility Code:

Module Overview for Biomedical Technician Workshop

Section 1: Spare Parts & Tools (direct towards the biomed maintenance staff)

Section 2: Storage & Layout (direct towards the biomed maintenance staff)

Section 3: Other Workshop Questions (direct towards the Head Biomed Engineer/technician in maintenance department)

Section 4: Maintenance & Repair (direct towards the biomed maintenance staff)

Appendix A: Workshop Layout Guidance

Facility Name:

Facility Code:

Section 1. Spare Parts and Tools

Instructions: Copy the Facility Code onto each page before the team begins to collect data. Please have all participants fill in their names and positions below, before completing the following forms in *Section 1. Spare Parts and Tools*.

Respondents	
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position

Facility Name:

Facility Code:

Section 1. Spare Parts and Tools

Instructions: Direct these questions to the biomed maintenance staff. Observe the items. The following questions will ask about the availability of various supplies. For the following questions you should first assess if the item is available today through observation. If no item is available, select **“not available today”**. If the item is available today and has not been out of stock in the last four weeks, select **“available and no stockout in last 4 weeks”**. If the item is available today, but there has been a stock out at any point in the last four weeks, select **“available today, but stockout in last 4 weeks”**.

Prompt: “I'd like to ask you a few questions on spare parts and tools.”

1A. Workshop

No.	Item	Response
1.01	Is there a dedicated workshop area?	☐ Yes ☐ No

1B. Personal Protective Equipment (PPE)

No.	Item	Response
1.02	What, if any, personal protective equipment is available in the workshop? (Read all response options aloud. Observe and select all that apply.)	☐ Overalls/jumpsuit ☐ Gloves (reusable) ☐ Face mask (dust/debris) ☐ Face mask (surgical) ☐ Face mask (N95) ☐ Face shield ☐ Goggles (infection prevention) ☐ Goggles (safety)

1C. Spare Parts

No.	Item	Response
General Spare Parts		
1.03	What general spare parts are available? (Read and observe)	
1.03a	Power cables	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.03b	9V batteries	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.03c	AA batteries	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.03d	AAA batteries	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.03e	Fuses	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
Glucometers		
1.04	What spare parts are available for glucometers? (Read and observe)	
1.04a	Batteries	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today

Facility Name:

Facility Code:

No.	Item	Response
Pulse Oximeters		
1.05	What spare parts are available for pulse oximeters? (<i>Read and observe</i>)	
1.05a	Neonatal sized pulse oximetry probe (reusable, clip)	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.05b	Neonatal sized pulse oximetry probe (reusable, wrap)	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.05c	Recharger for batteries	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
Radiant Warmers		
1.06	What spare parts are available for radiant warmers? (<i>Read and observe</i>)	
1.06a	Temperature probe	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.06b	Circuit board (main control)	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.06c	Heating element	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.06d	Power supply unit	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.06e	LCD	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
Incubators		
1.07	What spare parts are available for incubators? (<i>Read and observe</i>)	
1.07a	Circuit board (main control)	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.07b	Heating element	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.07c	Temperature probe	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.07d	Fan	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.07e	Control panel (membrane or LCD)	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today

Facility Name:

Facility Code:

No.	Item	Response
Oxygen Cylinders		
1.08	What spare parts are available for oxygen cylinders? <i>(Read and observe)</i>	
1.08a	Pressure regulator	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.08b	Stop valve	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.08c	Cap	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.08d	Pressure gauge	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.08e	Flowmeter	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.08f	Pigtail with safety wire	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
Oxygen Concentrators		
1.09	What spare parts are available for oxygen concentrators? <i>(Read and observe)</i>	
1.09a	Flowmeter	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.09b	Circuit board (control)	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.09c	Circuit board (oxygen monitoring)	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.09d	Circuit board/step down transformer (fan)	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.09e	Fan	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.09f	Solenoid valves (4-way)	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.09g	Solenoid valves (equalization)	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.09h	Molecular sieve beds	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks

Facility Name:

Facility Code:

No.	Item	Response
		☐ Not available today
1.09ii	Power supply unit	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.09j	Intake filter (internal, fine particle)	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.09k	Intake filter (external, gross particle)	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.09l	Reservoir tank	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.09m	Muffler with exhaust filter	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.09n	Compressor	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.09o	Compressor rebuild component	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.09p	Starting capacitor	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
Oxygen Flow Splitters		
1.10	What spare parts are available for oxygen flow splitters? (<i>Read and observe</i>)	
1.10a	Flowmeter	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
Suction Pumps (portable, electrical with accessories)		
1.11	What spare parts are available for suction pumps (portable, electrical with accessories)? (<i>Read and observe</i>)	
1.11a	Pump assembly	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.11b	Vacuum gauge	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.11c	Filter (bacterial)	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.11d	Collection reservoir	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today

Facility Name:

Facility Code:

No.	Item	Response
1.11e	Collection reservoir lid	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.11f	Power supply unit	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
Suction Pumps (manual, non-electric power dependent)		
1.12	What spare parts are available for suction pumps (manual, non-electric power dependent)? (Read and observe)	
1.12a	Piston assembly	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.12b	Vacuum gauge	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.12c	Filter (bacterial)	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.12d	Collection reservoir	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.12e	Collection reservoir lid	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
LED Phototherapy Lights		
1.13	What spare parts are available for LED phototherapy lights? (Read and observe)	
1.13a	Circuit board (control)	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.13b	Power supply unit	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.13c	LED driver/circuit/module	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.13d	Control panel (membrane or LCD)	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
Fluorescent Phototherapy Lights		
1.14	What spare parts are available for fluorescent phototherapy lights? (Read and observe)	
1.14a	Fluorescent bulb	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.14b	Power supply unit	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks

Facility Name:

Facility Code:

No.	Item	Response
		☐ Not available today
	CPAP Driver System (Standard or Bubble CPAP) with Accessories	
1.15	What spare parts are available for CPAP driver system (standard or bubble CPAP) with accessories? (<i>Read and observe</i>)	
1.15a	Pump assembly	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.15b	Filter (internal, pump)	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.15c	Flowmeter	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.15d	O-ring	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.15e	Power supply unit	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today

1D. Tools

No.	Item	Response
	What (if any) tools are available? (<i>Read and observe</i>)	
1.16	Digital Multi meter, digital, LED, with accessories; comparable or similar to Fluke brand piece	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.17	Penlight torch piece	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.18	Pocket magnifier piece	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.19	160 mm combination pliers (heavy duty) piece	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.20	Junior hack saw and blade piece	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.21	Hammer, soft-faced; 1 lb. piece	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.22	Hammer, ballpeen; 1 lb. piece	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today

Facility Name:

Facility Code:

No.	Item	Response
1.23	Soldering station with iron; 240 Vac, 300-500 watts	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.24	Soldering wire	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.25	Soldering tip cleaner (solution or wire sponge)	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.26	Desoldering pump or sucker	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.27	Solder braid wick	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.28	Imperial hex (Allen) key set	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.29	Tape measure; 3 meters; metal, metric piece	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.30	Metric hex (Allen) key set	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.31	Parallel tip screwdriver; 6 x 3/16 in. piece	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.32	Flared screwdriver; 4 x 1/4 in. piece	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.33	Diagonal cutter; 130 mm heavy duty piece	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.34	Straight tweezers piece	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.35	Polycarbonate brass trimmer set (7 items)	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.36	Insulated inspection screwdriver (165 mm) piece	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today

Facility Name:

Facility Code:

No.	Item	Response
1.37	Offset blade screwdriver (2 x 6 mm) pieces	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.38	Adjustable wire stripper/crimping tool combination; 0.25 to 0.8 mm piece	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.39	Terminal screwdriver piece	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.40	1.5 mm slot screwdriver piece	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.41	2.5 mm slot screwdriver piece	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.42	Combination spanners	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.43	Reversible screwdriver (No.1 and ¾ in. blades) each	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.44	Engineer's files; wooden handle; hand, square, warding, half round, round, and three-square in plastic wallet set	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.45	Torx (Tx 8, Tx 10, Tx 15 and Tx 20)	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.46	1.5 mm and 2.5 mm ball end drivers	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.47	Snap ring pliers	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.48	1.5 mm and 2.5 mm nut driver	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.49	Ratchet wrench and sockets	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.50	Bevelled cutter; 4 in. piece	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.51	Small snipe-nose pliers piece	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today

Facility Name:

Facility Code:

No.	Item	Response
1.52	Interchangeable Jeweler's screw drivers with 3 flats and 2 crosshead bits set	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.53	4 in. and 6 in. adjustable spanner each	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.54	150 mm utility pliers (heavy duty) piece	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today
1.55	Snap-off blade knife piece	☐ Available today and no stockout in last 4 weeks ☐ Available today, but stockout in last 4 weeks ☐ Not available today

Facility Name:

Facility Code:

Section 2. Storage and Layout

Instructions: Please have all participants fill in their names and positions below, before completing the following forms in *Section 2. Storage and Layout*.

Respondents	
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position

Facility Name:

Facility Code:

Section 2. Storage and Layout

Instructions: Direct these questions to the biomedical maintenance staff. For some questions, you will need to ask and observe the staff or specific items/spaces at the facility. If a response is unknown, write “unknown” or select “unknown” from the multiple-choice answers.

Prompt: “I’d like to ask you a few questions and observe the storage and layout of the workshop”.

Section 2A. Storage Location for New Equipment

No.	Item	Response	Skip To
2.01	Is there a storage location available for new equipment?	☐ Yes ☐ No	If “No”, skip to 2.07
2.02	Which of the following, if any, are available today in the storage location for new equipment? (Read all response options aloud. Observe and select all that apply.)		
2.02a	Pallet storage/appropriate racking	☐ Available today ☐ Not available today	—
2.02b	Temperature and relative humidity monitoring and recording systems	☐ Available today ☐ Not available today	—
2.02c	Fire extinguisher	☐ Available today ☐ Not available today	—
2.02d	Quality assurance system and record keeping	☐ Available today ☐ Not available today	—
2.02e	Controlled access (e.g., sign in sheet)	☐ Available today ☐ Not available today	—
2.02f	Lighting sufficient to view all racking areas	☐ Available today ☐ Not available today	—
2.02g	6 x 6 m or more free space for orderly packing, loading & unloading of equipment	☐ Available today ☐ Not available today	—
2.02h	Outside space for loading/unloading equipment	☐ Available today ☐ Not available today	—
2.02i	Transport structures (e.g. picking trolley)	☐ Available today ☐ Not available today	—
2.02j	First aid kit	☐ Available today ☐ Not available today	—
2.03	<i>For observation only:</i> Is the storage location for new equipment free and protected from insects, birds, and rodents?	☐ Yes ☐ No	—
2.04	<i>For observation only:</i> Is the storage location for new equipment visibly clean today?	☐ Yes ☐ No	—
2.05	<i>For observation only:</i> Is the storage location for new equipment secured?	☐ Yes ☐ No	—
2.06	<i>For observation only:</i> Is there ventilation in the storage location for new equipment?	☐ Yes ☐ No	—
2.07	What minimum documentation is available for each equipment order? (Observe documentation. Select all that apply.)	☐ None ☐ Date of receipt ☐ Name and description ☐ Quantity received ☐ Name and address of supplier	—

Facility Name:

Facility Code:

No.	Item	Response	Skip To
		☐ Batch numbers	
2.08	Is there a location for decommissioned equipment?	☐ Yes ☐ No	—

Section 2B. Workshop Space and Organization

No.	Item	Response
2.09	Does the maintenance workshop have an independent space?	☐ Yes ☐ No
2.10	What, if any, device sorting system is used? (Select all that apply)	☐ No sorting system used ☐ Ward ☐ Type ☐ Time received ☐ Other (specify below)

Section 2C. Workshop Layout and Infrastructure

Instructions: Please measure the dimensions of the workshop room(s) in cm and draw a basic outline on separate blank pieces of paper with the following designated sections and components.

If the workshop is made up of multiple rooms, please draw each room on a separate sheet of paper. The first diagram should be an overview of all the rooms and subsequent drawings should be detailed diagrams of each room. For each diagram, label what areas are in the diagram (e.g., overall layout, detailed diagram of storage areas, etc.).

Please take at least three photos of the workshop from different angles. If there are multiple rooms, please take photos of each room.

No.	Item	Notes
	Designated Sections	
2.11	Office space	
2.12	Service manuals section	
2.13	Tool storage section	
2.14	Spare parts storage section	
2.15	PPE storage section	
2.16	Devices in progress section	
2.17	Repaired devices section	
2.18	To be decommissioned section	
2.19	Equipment storage section	
	Components	
2.20	Repair bench	
2.21	Welding station	
2.22	Soldering station	
2.23	Shelving	
2.24	Racking	
2.25	Caging	
2.26	Doors	
2.27	Sinks	
2.28	Outlets	

Facility Name:

Facility Code:

Section 2D. Areas Shown in Photographs and Diagrams

Instructions: Please answer the following questions about the areas of the workshops shown in diagrams and photographs.

No.	Item	Notes
2.29	Which areas are in the photograph?	☐ Office space ☐ Service manuals section ☐ Tool storage ☐ Spare parts storage ☐ PPE storage ☐ Devices in progress section ☐ Repaired devices section ☐ To be decommissioned section ☐ Equipment storage section ☐ Components ☐ Repair bench ☐ Welding station ☐ Soldering station ☐ Shelving ☐ Racking ☐ Caging ☐ Other (<i>specify below</i>)

Facility Name:

Facility Code:

Section 3. Other Workshop Questions

Instructions: Please have all participants fill in their names and positions below, before completing the following forms in *Section 3. Other Workshop Questions*.

Respondents	
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position

Facility Name:

Facility Code:

Section 3. Other Workshop Questions

Instructions: Direct these questions to head biomedical engineer or technician of the maintenance department. For some questions, you will need to ask and observe the staff or specific items/spaces at the facility. If a response is unknown, write "unknown" or select "unknown" from the multiple-choice answers.

3A. Documentation

No.	Item	Response
3.01	What documentation is available in the workshop? (Select all that apply)	☐ None ☐ Corrective maintenance job cards ☐ Planned preventive maintenance job cards ☐ Stock cards ☐ Tool requisition register ☐ Spare parts requisition register ☐ Equipment/asset tags ☐ Decommissioning records ☐ Procurement records

3B. Planned Preventative Maintenance

No.	Item	Response	Skip To
3.02	Is there a Planned Preventive Maintenance System in use?	☐ Yes ☐ No	If "No", skip to 3.06
3.03	Which, if any, wards are covered by the Planned Preventive Maintenance System? (Select all that apply)	☐ None of the above ☐ Neonatal unit ☐ Postnatal unit ☐ Labour & delivery ward	
3.04	Are specific staff allocated to each ward? (Staff can be assigned to one ward or multiple wards)	☐ Yes ☐ No	
3.05	What is the frequency of scheduled preventive maintenance? (Select one)	☐ Biweekly ☐ Monthly ☐ Quarterly ☐ Annually ☐ Irregularly ☐ Other (specify below)	

Facility Name:

Facility Code:

3C. Corrective Maintenance

No.	Item	Response	Skip To
3.06	Is there a Corrective Maintenance System in use?	☐ Yes ☐ No	If "No", skip to 3.11
3.07	Which, if any, wards are covered by the Corrective Maintenance System? (Select all that apply)	☐ None of the above ☐ Neonatal unit ☐ Postnatal unit ☐ Labour & delivery ward	—
3.08	How are staff allocated for corrective maintenance? (Select one)	☐ By ward ☐ By availability ☐ Other (specify below)	—
			—
3.09	Is there an existing referral system in use to send devices offsite for maintenance?	☐ Yes ☐ No	—
3.10	Are corrective repairs documented?	☐ Yes ☐ No	—

3D. Decommissioning

No.	Item	Response	Skip To
3.11	Is there a decommissioning system in use?	☐ Yes ☐ No	If "No", skip to 3.14
3.12	Is there a clear, written procedure for decommissioning? (Observe written procedure)	☐ Yes ☐ No	—
3.13	Which, if any, of the following are included in the clear written procedure? (Read all response options aloud and select all that apply)	☐ None ☐ Decommission ☐ Decontaminate ☐ Eliminate	—
3.14	Is there an inventory of decommissioned equipment?	☐ Yes ☐ No	—
3.15	What, if any, challenges are there to decommissioning equipment? (Select all that apply)	☐ None ☐ Long process ☐ Too many approvers ☐ Lack of familiarity ☐ No plan for device replacement ☐ Other (specify below)	—
			—

Facility Name:

Facility Code:

3E. Equipment Leasing and Servicing Contracts

No.	Item	Response
3.16	Are equipment leasing and servicing contracts in use?	☐ Yes ☐ No ☐ Unknown
3.17	Is there a service provider assessment procedure?	☐ Yes ☐ No

Facility Name:

Facility Code:

3F. Electronic Maintenance System

No.	Item	Response
3.21	Is an electronic maintenance system in use?	☐ Yes ☐ No
3.21a	What is the name of the system?	Name:
3.22	What is covered by the electronic maintenance system? (Select all that apply)	☐ Equipment Inventory ☐ Spare Parts Inventory ☐ Planned Preventive Maintenance ☐ Corrective Maintenance Record ☐ Spare Parts Tracking ☐ Reporting ☐ Repair Library ☐ Decommissioning ☐ Other (specify below)

3G. Reporting

No.	Item	Response
3.23	Are monthly reports developed to report to any of the following audiences? (Select all that apply)	☐ None ☐ Hospital administration ☐ Regional/district health officers ☐ Ministry of Health (national) ☐ Unknown ☐ Other (specify below)
3.24	Are infrastructural issues (e.g., power or water interruptions) included in reports?	☐ Yes ☐ No ☐ Unknown
3.25	Are equipment maintenance issues included in reports?	☐ Yes ☐ No ☐ Unknown
3.26	For observation only: Are reports organized and available for review?	☐ Yes ☐ No

If reports are available for review, please write the date of the most recent report (yyyy/dd/mm):
 ____/____/____

Facility Name:

Facility Code:

3H. Human Resources

No.	Item	Response
3.27	Which, if any, of the following staff cadres are allocated to the maintenance department? (Select all that apply)	☐ Biomedical engineers ☐ Medical technicians/biomedics ☐ Spare parts requisition managers ☐ Tools requisition managers ☐ Electricians
3.28	How many of each staff type are currently employed in the maintenance department?	Biomedical engineers: Number:
		Medical technicians/biomedics: Number:
		Spare parts requisition managers: Number:
		Tools requisition managers: Number:
		Electricians: Number:
3.29	Are the biomedical engineers exclusively assigned to this facility?	☐ Yes ☐ No
3.30	Are the medical technicians/biomedics exclusively assigned to this facility?	☐ Yes ☐ No

Facility Name:

Facility Code:

3I. Staff Scheduling

Instructions: Someone is considered **physically present** if they are physically present anywhere at the facility, not just the neonatal ward. Someone is considered **on-call** if they can be called and reach the hospital within 30 minutes or if they are able to provide support needed by telephone. On-call **excludes the staff that are physically present at the time**, and only includes additional staff that have set times during which they could be called to the hospital if they are needed.

No.	Which of the following time(s) is the facility staffed by medical technicians/biomedics or engineers?	Response		
	Regular Schedule (not on-call staff)	Is the biomed staff physically present?		If yes, how many?
3.31	Weekday (e.g. Monday – Friday) during the day	☐ Yes	☐ No	Number:
3.32	Weekday (e.g. Monday – Friday) during the night	☐ Yes	☐ No	Number:
3.33	Weekend (e.g. Saturday-Sunday) during the day	☐ Yes	☐ No	Number:
3.34	Weekend (e.g. Saturday-Sunday) during the night	☐ Yes	☐ No	Number:
3.35	Last night	☐ Yes	☐ No	Number:
3.36	Today	☐ Yes	☐ No	Number:
	On-Call Schedule	Is biomed staff on-call?		If yes, how many?
3.37	Weekday (e.g. Monday – Friday) during the day	☐ Yes	☐ No	Number:
3.38	Weekday (e.g. Monday – Friday) during the night	☐ Yes	☐ No	Number:
3.39	Weekend (e.g. Saturday-Sunday) during the day	☐ Yes	☐ No	Number:
3.40	Weekend (e.g. Saturday-Sunday) during the night	☐ Yes	☐ No	Number:
3.41	Last night	☐ Yes	☐ No	Number:
3.42	Today	☐ Yes	☐ No	Number:

3J. Other

No.	Item	Response
3.43	Is there a procurement plan in place for medical devices?	☐ Yes ☐ No
3.44	Are there guidelines or a system for receiving equipment donations?	☐ Yes ☐ No
3.45	Which, if any, of the following meetings include biomedical or maintenance staff? (Select all that apply)	☐ None ☐ Management meetings ☐ Budget planning ☐ Procurement planning ☐ QI meetings

Facility Name:

Facility Code:

Section 4. Maintenance and Repair

Instructions: Please have all participants fill in their names and positions below, before completing the following forms in *Section 4. Maintenance and Repair*.

Respondents	
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position

Facility Name:

Facility Code:

Section 4. Maintenance and Repair

Instructions: Direct these questions to head biomedical engineer or technician of the maintenance department. For some questions, you will need to ask and observe the staff or specific items/spaces at the facility. If a response is unknown, write “unknown” or select “unknown” from the multiple-choice answers.

4A. In the Neonatal Unit

No.	Item	Response
4.01	<i>For observation only:</i> Is there an inventory list of all the equipment in this unit?	☐ Yes ☐ No
4.02	Who employs the technician?	☐ Hospital ☐ Ministry of Health ☐ Distributor ☐ Manufacturer ☐ Unknown ☐ Other (specify below)
4.03	Is repair of medical equipment free for the facility?	☐ Yes ☐ No ☐ Unknown
4.04	Who primarily funds the spare parts when equipment breaks? (Select all that apply)	☐ Hospital ☐ Government ☐ Donation from NGO ☐ Private purchase ☐ Unknown ☐ Other (specify below)
4.05	What do you think is the biggest barrier to repairing equipment? (Select one)	☐ Insufficient number of technicians ☐ Lack of training for technician ☐ Cost of repair ☐ Lack of spare parts ☐ Lack of tools for repair ☐ Other (specify below)

Facility Name:

Facility Code:

4B. Biomedical Services

Instructions: 'On the job training' is defined as any training undertaken while you are doing the job, including mentorship, informal training, shadowing and side-by-side work. 'A course' is defined as any formal training undertaken during your employment period, but not while you are at work. This could include any offsite or onsite training course.

No.	Item	Response
4.06	Repairing CPAP Device	
4.06a	Does your facility provide this service?	☐ Yes ☐ No
4.06b	Were you trained during pre-service training on how to provide this service?	☐ Yes ☐ No
4.06c	Have you ever received a course on how to provide this service?	☐ Yes ☐ No
4.06d	Have you ever received on the job training on how to provide this service?	☐ Yes ☐ No
4.07	Repairing Oxygen Concentrators	
4.07a	Does your facility provide this service?	☐ Yes ☐ No
4.07b	Were you trained during pre-service training on how to provide this service?	☐ Yes ☐ No
4.07c	Have you ever received a course on how to provide this service?	☐ Yes ☐ No
4.07d	Have you ever received on the job training on how to provide this service?	☐ Yes ☐ No
4.08	Providing Planned Preventive Maintenance on Oxygen Concentrators	
4.08a	Does your facility provide this service?	☐ Yes ☐ No
4.08b	Were you trained during pre-service training on how to provide this service?	☐ Yes ☐ No
4.08c	Have you ever received a course on how to provide this service?	☐ Yes ☐ No
4.08d	Have you ever received on the job training on how to provide this service?	☐ Yes ☐ No
4.09	Repairing Phototherapy Lights	
4.09a	Does your facility provide this service?	☐ Yes ☐ No
4.09b	Were you trained during pre-service training on how to provide this service?	☐ Yes ☐ No
4.09c	Have you ever received a course on how to provide this service?	☐ Yes ☐ No
4.09d	Have you ever received on the job training on how to provide this service?	☐ Yes ☐ No

Facility Name:

Facility Code:

No.	Item	Response
4.10	Providing Planned Preventive Maintenance on Phototherapy Lights	
4.10a	Does your facility provide this service?	☐ Yes ☐ No
4.10b	Were you trained during pre-service training on how to provide this service?	☐ Yes ☐ No
4.10c	Have you ever received a course on how to provide this service?	☐ Yes ☐ No
4.10d	Have you ever received on the job training on how to provide this service?	☐ Yes ☐ No

Facility Name:

Facility Code:

Appendix A. Workshop Layout Guidance

Instructions: Draw the floorplan of the biomedical workshop, including separate rooms (with a description of the purposes of each of the rooms), equipment, windows, doors & other infrastructural items (as shown in legend below). Draw the overall layout showing the relation of the rooms to each other, and each room on a separate page with the detailed items listed below.

Legend



Door; draw on floor with arc following the way that door opens



Door; draw on wall in door's location



Window; draw to size and shape of windows. Draw all connected louvers as one window, but louvers separated by wood frames as separate windows



Sockets/power sources with 1 plug



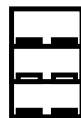
Sockets/power sources with 2 plugs



repair bench



Racking, draw on floor



Racking, draw on wall



Caging, draw on floor



Caging, draw on wall



Sink



Welding station